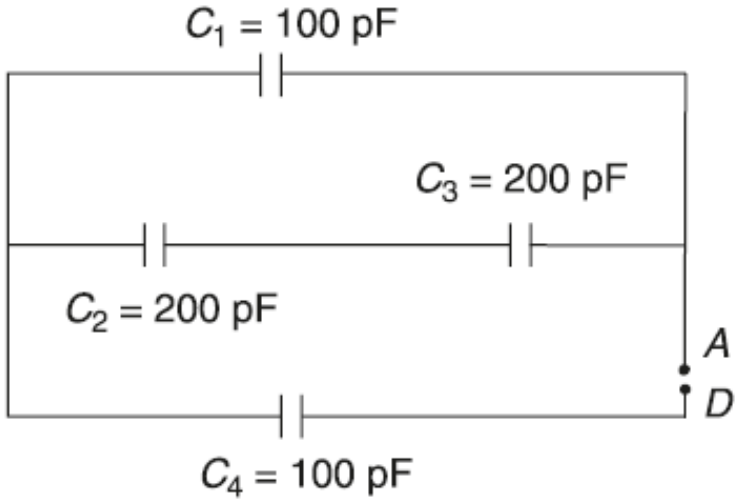
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

What is the equivalent capacitance between A and D of the network shown in Fig.?



(A) 200pF

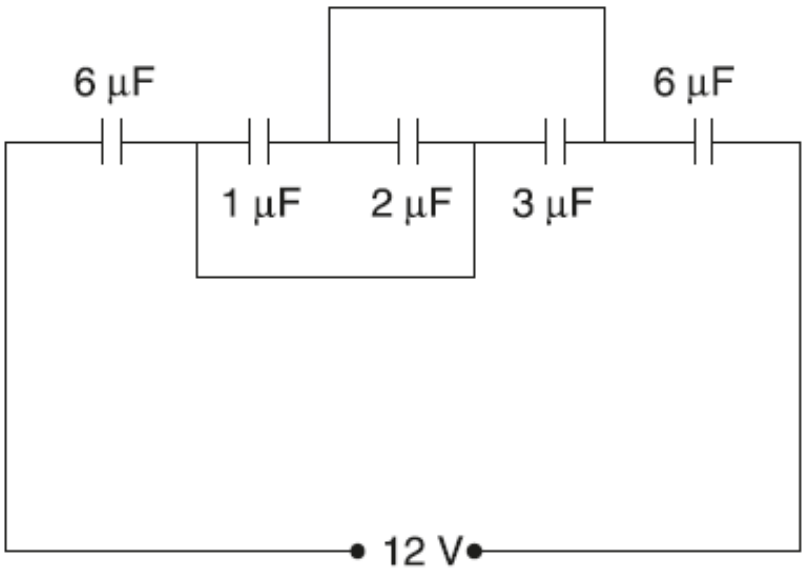
(B) 100pF

(C) $\frac{200}{3}\text{pF}$

(D) 50pF

Q.2

Figure shows five capacitors connected across a $12\ \text{V}$ power supply. What is the charge on the $2\mu\text{F}$ capacitor?



(A) $6\mu\text{C}$

(B) $8\mu\text{C}$

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(C) $10\mu C$

(D) $12\mu C$

Q.3

In a parallel-plate capacitor of capacitance C, a metal sheet is inserted between the plates parallel to them. The thickness of the sheet is half of the separation between the plate The capacitance now becomes

(A) $4C$

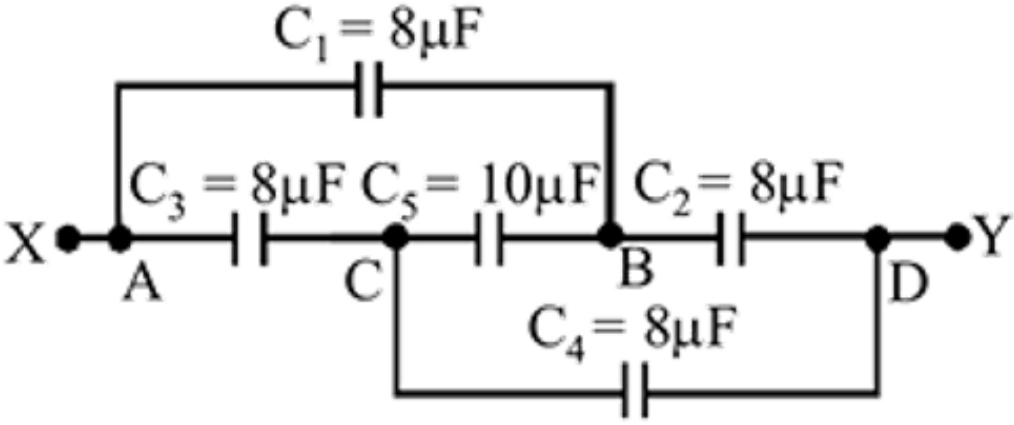
(B) $2C$

(C) $C/2$

(D) $C/4$

Q.4

What is the effective capacitance between points X and Y?



(A) $8\mu F$

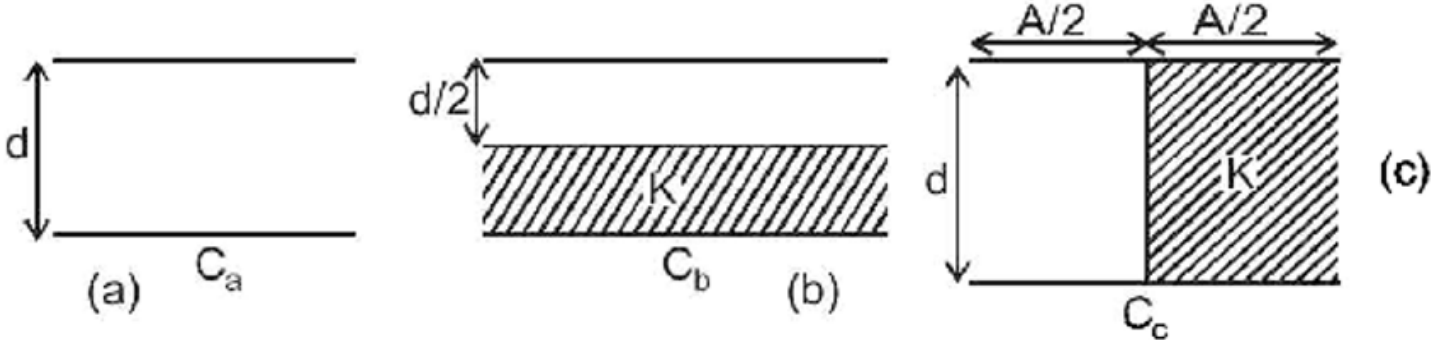
(B) $9\mu F$

(C) $10\mu F$

(D) $12\mu F$

Q.5

The capacitance of a parallel plate capacitor is $\mathrm{C_a}$ (Fig. a). A dielectric of dielectric constant K is inserted as shown in fig (b) and (c) then



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(A) $\mathrm{Cb} < \mathrm{Ca}$ and $\mathrm{Cc} > \mathrm{Ca}$

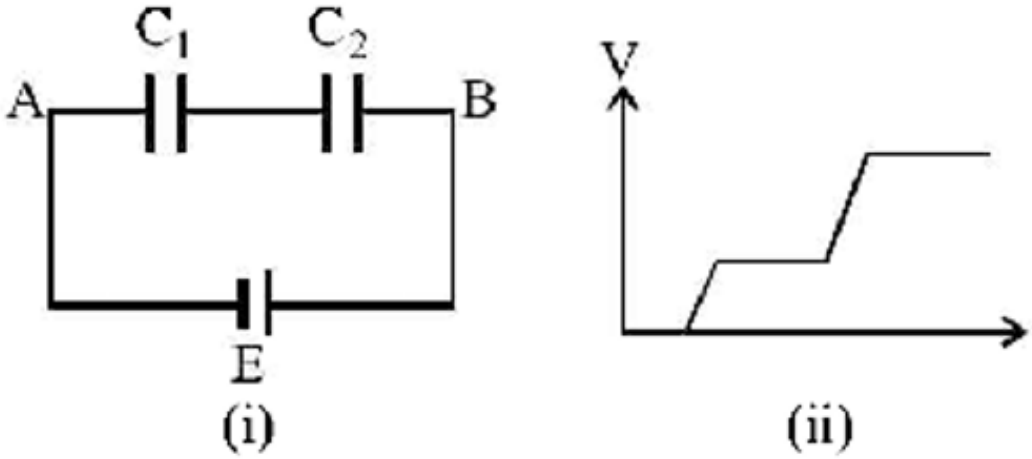
(B) $\mathrm{C_c} < \mathrm{C_a}$ while $\mathrm{C_b} > \mathrm{C_a}$

(C) both $C_b, C_c < C_a$

(D) $C_a = C_b = C_c$

Q.6

Figure (i) shows two capacitors connected in series and connected by a battery. The graph (ii) shows the variation of potential as one moves from left to right on the branch AB containing the capacitors. Then



(A) $C_1 = C_2$

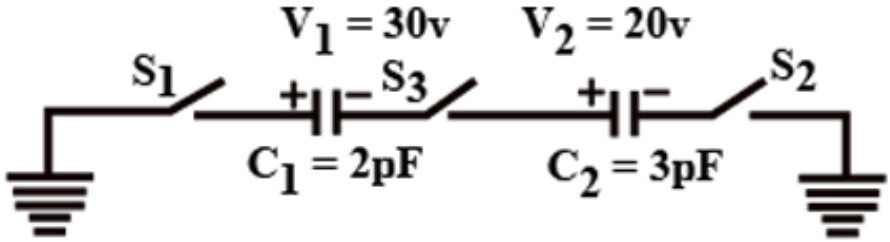
(B) $\mathrm{C_1} < \mathrm{C_2}$

(C) $\mathrm{C_1} > \mathrm{C_2}$

(D) $\mathrm{C_1}$ and $\mathrm{C_2}$ cannot be compared

Q.7

For the circuit shown in figure, which of the following statements is true?



(A) With $\mathrm{S_1}$ closed $\mathrm{V_1} = 15\ \mathrm{V},\ \mathrm{V_2} = 20\ \mathrm{V}$

(B) With $\mathrm{S_3}$ closed $\mathrm{V_1} = \mathrm{V_2} = 25\ \mathrm{V}$

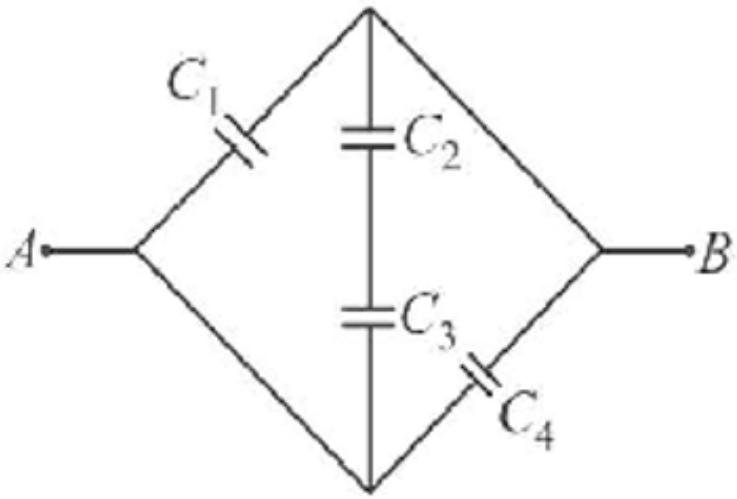
(C) With $\mathrm{S_1}$ and $\mathrm{S_2}$ closed $\mathrm{V_1} = \mathrm{V_2} = 0$

(D) With $\mathrm{S_1}$ and $\mathrm{S_3}$ closed, $\mathrm{V_1} = 30\ \mathrm{V},\ \mathrm{V_2} = 20\ \mathrm{V}$

Q.8

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In a given network the equivalent capacitance between A and B is $[C_1 = C_4 = 1\mu\text{F}, C_2 = C_3 = 2\mu\text{F}]$



(A) $3\mu F$

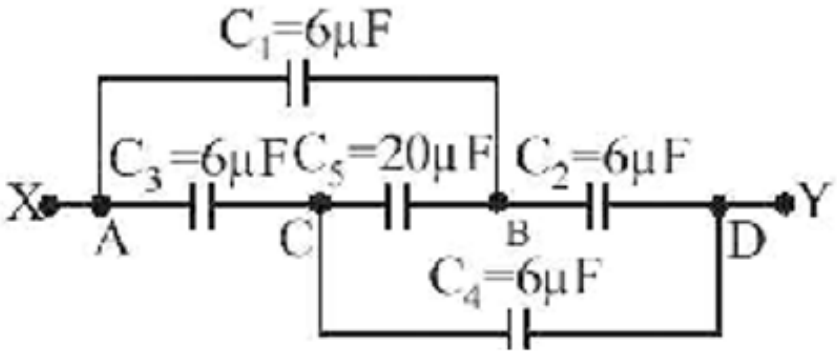
(B) $6\mu F$

(C) $4.5\mu F$

(D) $2.5\mu F$

Q.9

What is the effective capacitance (in μF) between points X and Y?



(A) 5

(B) 6

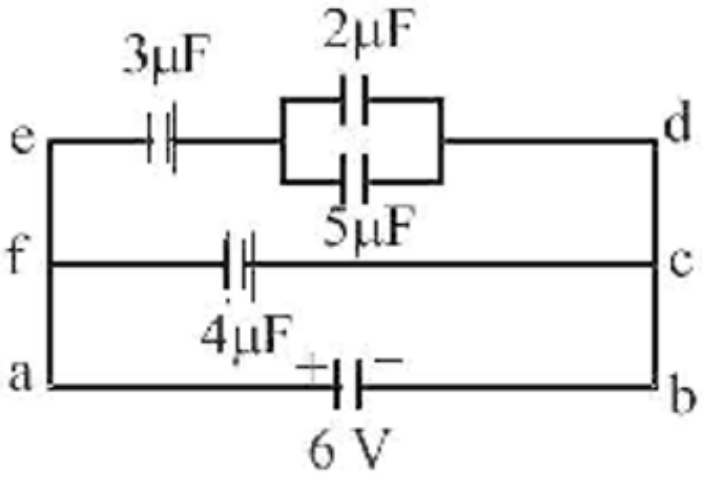
(C) 7

(D) 8

Q.10

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In the circuit given below, what is the charge in μC, on the capacitor having $5\mu F$?



(A) 6

(B) 7

(C) 8

(D) 9

Q.11

Two large conducting parallel plates having equal charge Q are placed very close to each other and distance between the plates is d. Find the work done by external agent to increase the separation between plates by d (area of plate is A)



(A) $\frac{Q^2 d}{2A\varepsilon_0}$

(B) $-\frac{Q^2 d}{2A\varepsilon_0}$

(C) $-\frac{3Q^2 d}{2A\varepsilon_0}$

(D) $-\frac{Q^2 d}{A\varepsilon_0}$

Q.12

A positive charge q is given to each plate of a parallel plate air capacitor having plate area A and plate separation d, then

(A) since both the plates are identically charged, therefore, capacitance becomes equal to zero

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(B) energy stored in the space between the capacitor plates is equal to $\frac{q^2}{\varepsilon_0 A^2}$

(C) no charge appears on inner surface of the plates

(D) potential difference between the plates is equal to $\frac{2qd}{\varepsilon_0 A}$

Q.13

A parallel plate capacitor of capacitance C is charged with a battery of potential V. The battery is then disconnected and electromagnetic waves are incident on negative plate of the capacitor. As a result the negative plate starts emitting electrons towards the positive plate. The current which flows between the plates remains constant till time t_1 and then starts decreasing. The potential difference between the plates at time t_1 is (Assume plates of capacitor are close to each other)

(A) V

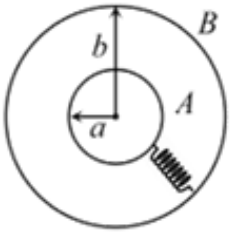
(B) $V/2$

(C) $2\ \mathrm{V}$

(D) zero

Q.14

Two spherical conductors A and B of radii a and b $(b > a)$ are placed concentrically in air. The two are connected by a copper wire as shown in figure. Then the equivalent capacitance of the system is



(A) $\frac{4\pi\varepsilon_0 ab}{b-a}$

(B) $4\pi\varepsilon_0(a+b)$

(C) $4\pi\varepsilon_0 b$

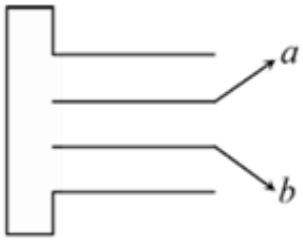
(D) $4\pi\varepsilon_0 a$

Q.15

Four metallic plates, each with a surface area of one side A, are placed at a distance d from each other. The

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plates are connected as shown in the figure. Then the capacitance of the system between a and b is



(A) $\frac{3\varepsilon_0 A}{d}$

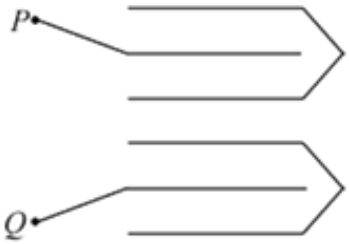
(B) $\frac{2\varepsilon_0 A}{d}$

(C) $\frac{2\varepsilon_0 A}{3d}$

(D) $\frac{3\varepsilon_0 A}{2d}$

Q.16

In the arrangement shown all the plates have equal area A and spacing d between them. The equivalent capacitance between point P and Q will be,



(A) $\frac{\varepsilon_0 A}{d}$

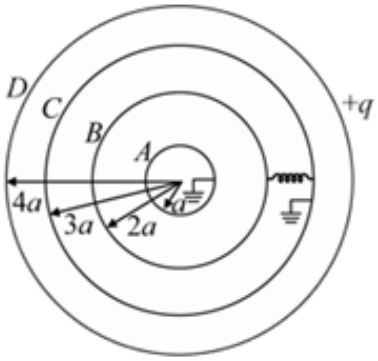
(B) $\frac{\varepsilon_0 A}{3d}$

(C) $\frac{\varepsilon_0 A}{2d}$

(D) $\frac{2\varepsilon_0 A}{d}$

Q.17

Four concentric conducting shells A, B, C and D are arranged as shown in figure. A charge $+q$ is given to the outer most shell D. Shell B and C are connected by conducting wire while Shell A and C is earthed. The equivalent capacitance of system is



(A) $64\pi\varepsilon_0 a$

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(B) $16\pi\varepsilon_0 a$

(C) $4\pi\varepsilon_0 a$

(D) none of these

Q.18

When air in a capacitor is replaced by a medium of dielectric constant K, the capacitance

(A) decreases K times

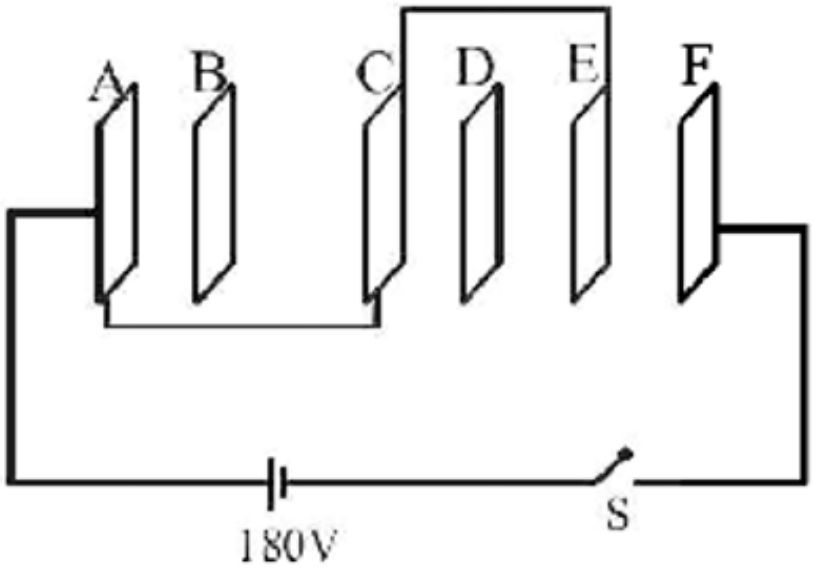
(B) increases K times

(C) increases K^2 times

(D) remains constant

Q.19

$\mathrm{A, B, C, D, E, F}$ are conduct- ing plates each of area A and any two consecutive plates separated by a distance d. The net energy stored in the system after the switch S is closed is



(A) $\frac{3\varepsilon_0\,\mathrm{A}}{2\,\mathrm{d}}\,\mathrm{V}^2$

(B) $\frac{5\varepsilon_0\,\mathrm{A}}{12\,\mathrm{d}}\,\mathrm{V}^2$

(C) $\frac{\varepsilon_0\,\mathrm{A}}{2\,\mathrm{d}}\,\mathrm{V}^2$

(D) $\frac{\varepsilon_0\,\mathrm{A}}{\mathrm{d}}\,\mathrm{V}^2$

Q.20

The space between the plates of a parallel plate capacitor is filled with a 'dielectric' whose 'dielectric constant' varies with distance as per the relation:

$\mathrm{K(x)} = \mathrm{K_o} + \lambda \mathrm{x} (\lambda = \text{a constant})$

The capacitance C, of the capacitor, would be related to its vacuum capacitance $\mathrm{C_o}$ for the relation :

(A) $\mathrm{C} = \frac{\lambda \mathrm{d}}{\ln(1+\mathrm{K_o}\lambda \mathrm{d})}\mathrm{C_o}$

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(B) $C = \frac{\lambda}{d.\ln(1+K_o\lambda d)}C_o$

(C) $C = \frac{\lambda d}{\ln(1+\lambda d/K_o)}C_o$

(D) $C = \frac{\lambda}{d\cdot\ln(1+K_o/\lambda d)}C_o$

Answers & Solutions

Q.1 Answer:

$\frac{200}{3}$pF

Solution:

The series combination of C_2 and C_3 is equivalent to a capacitance C' given by

$\frac{1}{C'} = \frac{1}{C_2} + \frac{1}{C_3}$

or $C' = \frac{C_2C_3}{C_2+C_3} = \frac{200\times200}{200+200} = 100\text{pF}$

Therefore the circuit reduces to the one shown in Fig. 11.93(a). The equivalent capacitance between points A and B is

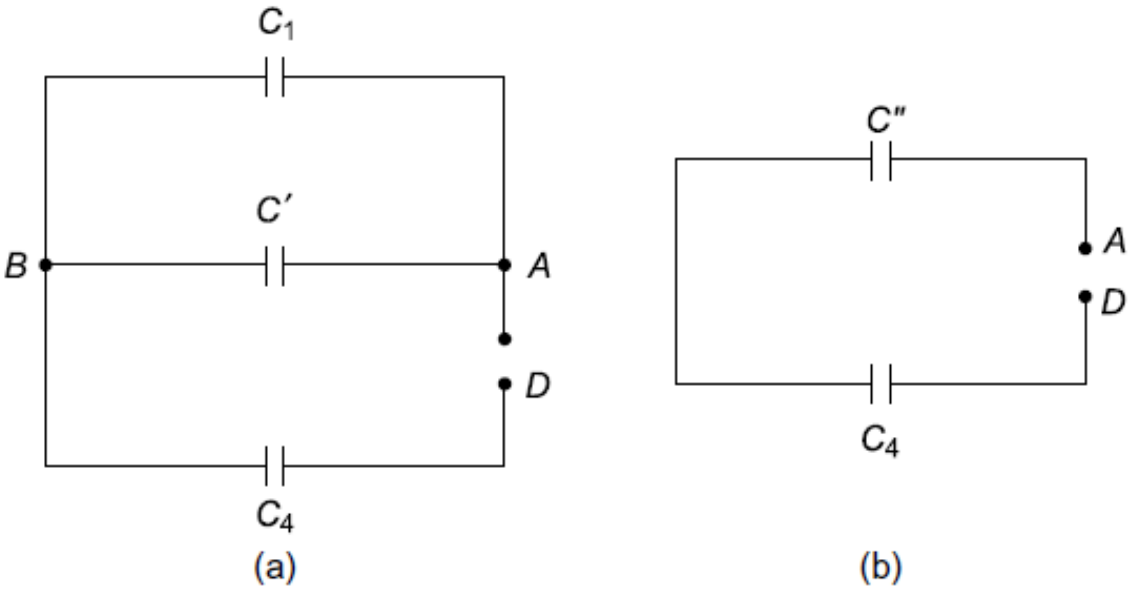
$C'' = C_1 + C' = 100 + 100 = 200\text{pF}$

The circuit may be further simplified to that in Fig. $11.93(\text{ b})$. The equivalent capacitance C of the entire network, i.e., between points A and D, is now that of the series combination of C'' and C_4.

Thus

$$\frac{1}{C} = \frac{1}{C''} + \frac{1}{C^4} = \frac{1}{200} + \frac{1}{100}$$
$$= \frac{3}{200} \quad \text{or} \quad C = \frac{200}{3}\text{pF}$$

Hence the correct choice is (c).



Q.2 Answer:

$8\mu\text{C}$

Solution:

Capacitors $1\mu\text{F}, 2\mu\text{F}$ and $3\mu\text{F}$ are in parallel, their total capacitance is $6\mu\text{F}$. Thus, we have three capacitors in series each of capacitance $6\mu\text{F}$ across the 12 V power supply. So, the potential drop

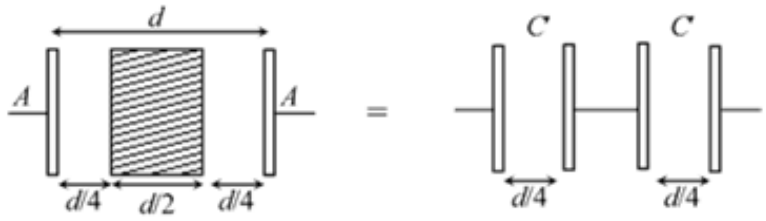
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across each is $12/3 = 4$ V. This is also the potential across 1μF capacitor and 2μF capacitor and 3μF capacitor, because they are in parallel. Therefore, charge on 2μF capacitor $= 2\mu\text{F} \times 4\text{ V} = 8\mu\text{C}$. Hence the correct choice is (b).

Q.3 Answer:

$2C$

Solution:



Before the metal sheet is inserted, $C = \frac{\varepsilon_0 A}{d}$
After the sheet is inserted, the system is equivalent to two capacitors in series, each of capacitance $C = \frac{\varepsilon_0 A}{(d/4)} = 4C$.
The equivalent capacity is now $2C$.

Q.4 Answer:

$8\mu\text{F}$

Solution:

By symmetry $\text{C}_5 = 10\mu\text{F}$ Capacitor between B and C can be removed so, now $C_{eq} = \frac{8\times8}{8+8} + \frac{8\times8}{8+8} = 8\mu\text{F}$

Q.5 Answer:

$\text{Cb} < \text{Ca}$ and $\text{Cc} > \text{Ca}$

Solution:

$\text{C}_\text{a} = \frac{\epsilon_0\,\text{A}}{\text{d}}$ **and** $\text{C}_\text{b} = \frac{\epsilon_0\,\text{A}}{\frac{\text{d}}{2}+\frac{\text{d}}{2\,\text{K}}} = \frac{2\varepsilon 0\text{AK}}{\text{d}(1+\text{K})} = \text{Ca}(\text{K}/1+\text{K})$
and $\text{C}_\text{c} = \frac{\epsilon_0\frac{\text{A}}{2}}{\text{d}} + \frac{\epsilon_0\frac{\text{A}}{2}\,\text{K}}{\text{d}} = \frac{\epsilon_0\,\text{A}}{2\,\text{d}}(1+\text{K})$
or $\text{Cb} = \text{Ca}/(1+1/\text{K})$
Here $(1+1/K) > 1$ irrespective of the value of K
So, $\text{Cb} < \text{Ca}$
or $\text{C}_\text{c} = \frac{\epsilon_0\,\text{A}}{\text{d}}\frac{1+\text{K}}{2} > \text{C}_\text{a}$ $\quad\therefore \text{Cb} < \text{Ca}$ **and** $\text{Cc} > \text{Ca}$

Q.6 Answer:

$\text{C}_1 > \text{C}_2$

Solution:

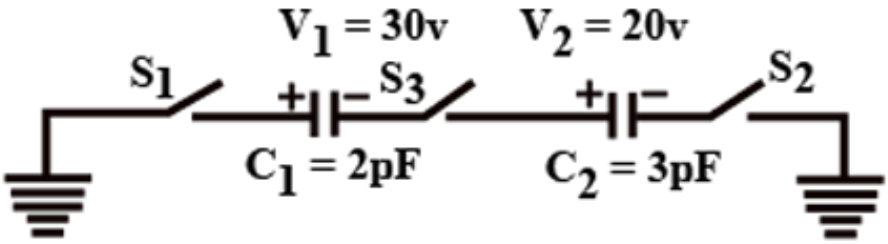
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Since, potential difference across C_2 is greater than C_1. $\Rightarrow C_1 > C_2$ $\left[\because V = \frac{q}{C}\right.$ and q is same in series

Q.7 Answer:

With S_1 and S_3 closed, $V_1 = 30\ V,\ V_2 = 20\ V$

Solution:



If we close any switch there will be no change in any potential because,

If S_1 will close C_1 will connect to ground but no negative charges will from ground to capacitor because of negatively charged plate of C_1.

If S_2 will close same will happen as with the case of S_1 closed.

But if S_3 is closed.

Charge on negative plate of $C_1 = 2 \times 10^{-12} \times 30$

$= 60 \times 10^{-12}$

Charge on positive plate of $C_2 = 3 \times 10^{-12} \times 20$

$= 60 \times 10^{-12}$

$\therefore$ No charge redistribution occur because equilent positive and negative charge is present on C_1 and C_2 respectively.

$\therefore$ U_1 and U_2 will remainsame.

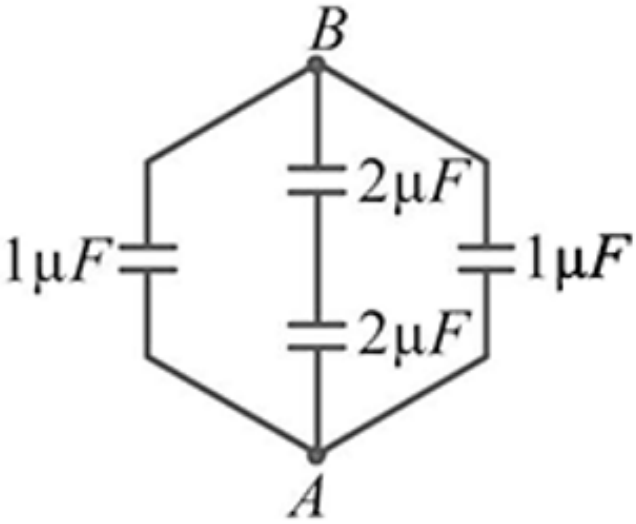
i.e $V_1 = 30v$ **and** $V_2 = 20v$

Q.8 Answer:

$3\mu F$

Solution:

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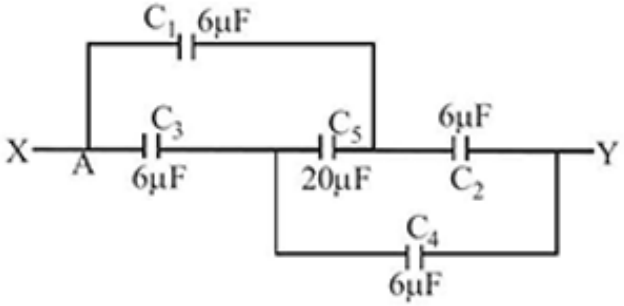


The equivalent circuit is shown in figure. $C_{AB} = 3\mu F$

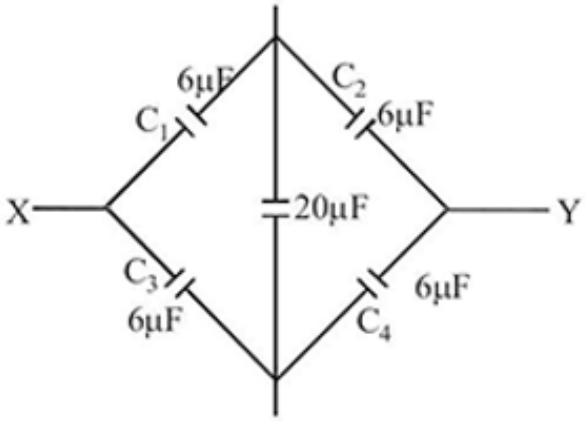
Q.9 Answer:

6

Solution:



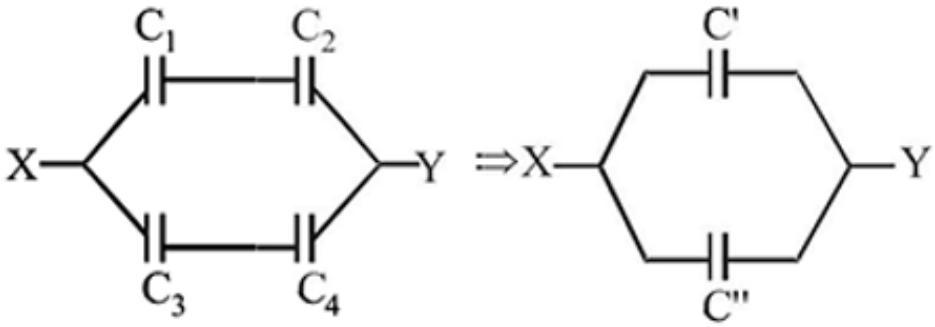
Equivalent circuit,



Here, $\frac{C_1}{C_3} = \frac{C_2}{C_4}$

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Hence, no charge will flow through $20\mu\mathrm{F}$



C_1 and C_2 are in series, also C_3 and C_4 are in series. Hence, $C = 3\mu F, C'' = 3\mu F$
C' and C'' are in parallel.

Q.10 Answer:

9

Solution:

Potential difference across the branch de is $6\ \mathrm{V}$. Net capacitance of de branch is $2.1\mu\mathrm{F}$ So, $\mathrm{q} = \mathrm{CV}$
$\Rightarrow \mathrm{q} = 2.1 \times 6\mu\mathrm{C}$
$\Rightarrow \mathrm{q} = 12.6\mu\mathrm{C}$
Potential across $3\mu\mathrm{F}$ capacitance is $\mathrm{V} = \frac{12.6}{3} = 4.2\mathrm{volt}$
Potential across 2 and 5 combination in parallel is $6 - 4.2 = 1.8\ \mathrm{V}$
So, $q' = (1.8)(5) = 9\mu\mathrm{C}$

Q.11 Answer:

$-\frac{Q^2 d}{2A\varepsilon_0}$

Solution:

Force on first plate due to second plate $\left(\frac{Q}{2A\varepsilon_0}\right) Q$
$W_{\text{ext}} = -W_{\text{electric field}} = -\frac{Q^2 d}{2A\varepsilon_0}$



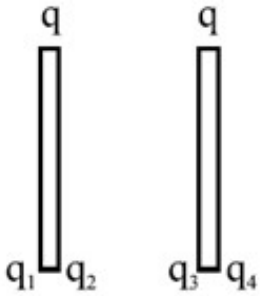
Q.12 Answer:

no charge appears on inner surface of the plates

Solution:

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Capacitance does not depend on charge on it, $C = \frac{\varepsilon_0 A}{d}$



In final charge distribution, Charge on outer plates $q_1 = q_4 = q$ **and inner plates** $q_2 = q_3 = 0$

As charge on inner plates are zero so there is no field between the plates and no potential difference and no energy is stored.

Q.13 Answer:

zero

Solution:

The current starts decreasing when the potential difference between the plates becomes zero.

Q.14 Answer:

$4\pi\varepsilon_0 b$

Solution:

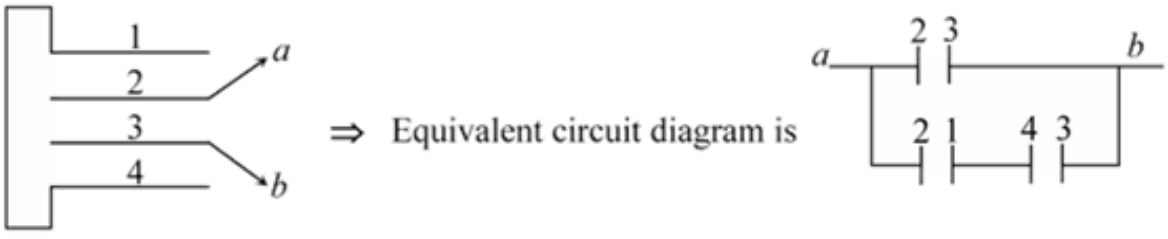
All the charge given to inner sphere will pass on to the outer one.

So, capacitor of that outer one $= 4\pi\varepsilon_0 b$**.**

Q.15 Answer:

$\frac{3\varepsilon_0 A}{2d}$

Solution:



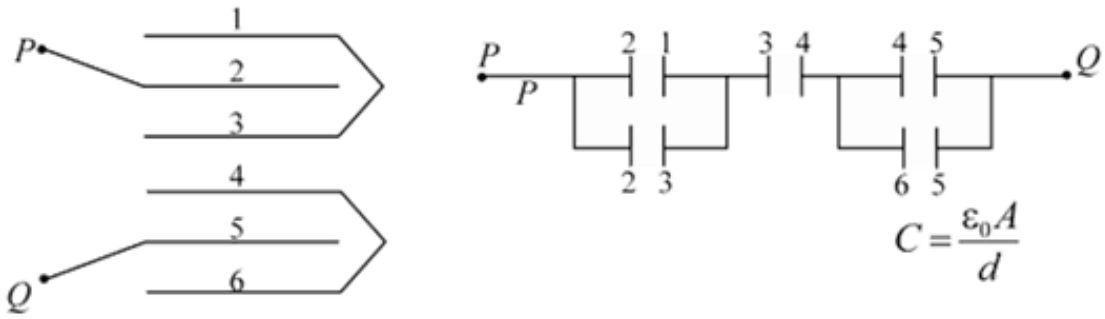
$C_{eq} = \frac{3}{2}\frac{\varepsilon_0 A}{d}$

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Q.16 Answer:

$\frac{\varepsilon_0 A}{2d}$

Solution:



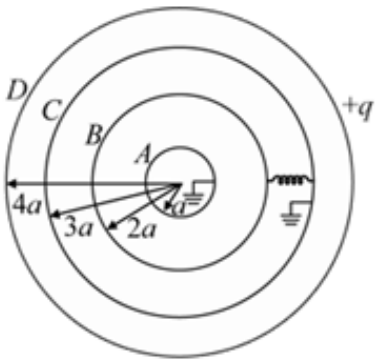
$\frac{1}{C_{\text{eq}}} = \frac{1}{2C} + \frac{1}{C} + \frac{1}{2C} = \frac{4}{2C}$

$C_{\text{eq}} = \frac{C}{2} = \frac{\varepsilon_0 A}{2d}$

Q.17 Answer:

$64\pi\varepsilon_0 a$

Solution:



The outer shell of circle D has $+q$ charge.

Circle A and C has no charge in it due to earthing.

So circle B has $-q$ charge in it.

We know that the capacitance between two concentric Circles of radius r_1 & r_2 is

$c = \frac{4\pi\epsilon_0 r_1 r_2}{r_1 - r_2}$

Here we take only D and B circle, as these two have charge in t.

So the equivalent capacitance is

$$c = \frac{4\pi\epsilon_0 4a \times 2a}{4a - 2a}$$

$$c = \frac{4\pi\epsilon_0}{2a} 2a \times 4a$$

$$c = 16\pi\epsilon_0 a$$

Therefore the equivalent capacitance of the system is $16\pi\epsilon_0 a$.

Option (B) is Correct.

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Q.18 Answer:

increases K times

Solution:

$C_{\text{medium}} = K \times C_{\text{air}}$

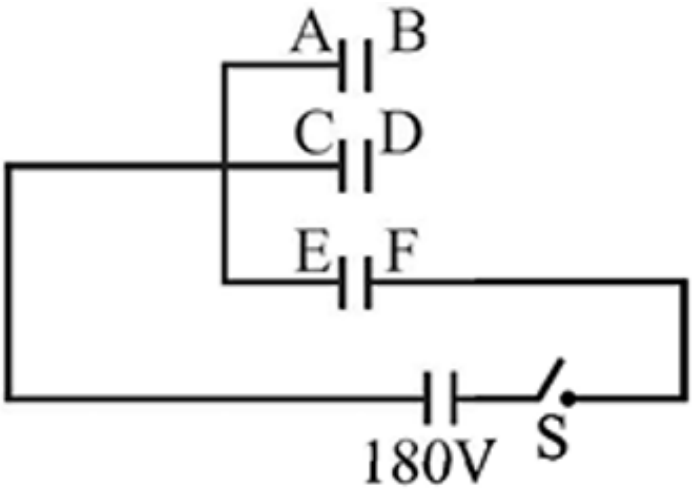
Q.19 Answer:

$\frac{\varepsilon_0 \, \mathrm{A}}{2 \, \mathrm{d}} \, \mathrm{V}^2$

Solution:

$\mathrm{C_{eff}} = \frac{\varepsilon_0 \, \mathrm{A}}{\mathrm{d}}$ **since effective capacitance between plates** A **and** E **is zero.**

$$\therefore \quad \mathrm{U} = \frac{1}{2}\mathrm{CV}^2 = \frac{\varepsilon_0 \, \mathrm{A}}{2 \, \mathrm{d}} \, \mathrm{V}^2$$



Q.20 Answer:

$\mathrm{C} = \frac{\lambda \mathrm{d}}{\ln(1+\lambda \mathrm{d}/\mathrm{K_o})} \mathrm{C_o}$

Solution:

The value of dielectric constant is given as, $\mathrm{K} = \mathrm{K_0} + \lambda \mathrm{x}$

And, $\mathrm{V} = \int_0^{\mathrm{d}} \mathrm{E} \mathrm{dr}$

$\mathrm{V} = \int_0^{\mathrm{d}} \frac{\sigma}{\mathrm{K}} \mathrm{dx}$

$= \sigma \int_0^{\mathrm{d}} \frac{1}{(\mathrm{K_0}+\lambda \mathrm{x})} \mathrm{dx}$

$= \frac{\sigma}{\lambda} \left[\ln(\mathrm{K_0} + \lambda \mathrm{d}) - \ln \mathrm{K_0}\right]$

$= \frac{\sigma}{\lambda} \ln\left(1 + \frac{\lambda \mathrm{d}}{\mathrm{K_0}}\right)$

Now it is given that capacitance of vacuum $= \mathrm{C_0}$**. Thus,** $\mathrm{C} = \frac{\mathrm{Q}}{\mathrm{V}}$

$= \frac{\sigma . \mathrm{S}}{\mathrm{v}}$ **(Let surface area of plates** $= \mathrm{s}$**)**

$= \frac{\sigma . \mathrm{s}}{\frac{\sigma}{\lambda} \ln\left(1+\frac{\lambda \mathrm{d}}{\mathrm{K_0}}\right)}$

$= \mathrm{s}\lambda \cdot \frac{\mathrm{d}}{\mathrm{d}} \frac{1}{\ln\left(1+\frac{\lambda \mathrm{d}}{\mathrm{K_0}}\right)}$ ($\because$ **in vacuum** $\varepsilon_0 = 1$)

$\mathrm{C} = \frac{\lambda \mathrm{d}}{\ln\left(1+\frac{\lambda \mathrm{d}}{\mathrm{K_0}}\right)} \cdot \mathrm{C_0} \quad$ (**here,** $\mathrm{C_0} = \frac{\mathrm{s}}{\mathrm{d}}$)

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